SCIENCE, TECHNOLOGY AND SOCIETY

Deals with the interactions between **science and technology and social**, **cultural, political and economic context** that shape and are shaped by them. (CMO No. 20, series of 2013).

There are varied and different ways to define **SCIENCE AND TECHNOLOGY** such that its connection and relevance to society can be actualized.

• Interdisciplinary study of the interaction of science and technology with society and culture.

• The realization that discoveries and inventions are shaped by historical forces and in turn influence values, aspirations, events, and institutions, thus shaping the course of history

Science is…

•**Content**: Body of organized knowledge about nature

•**Method**: of obtaining that knowledge, experiment, observation, hypothesis, theory, law

•**Attitude**: organized and systematic skepticism

•**Goals**: Explanation, understanding, prediction, control

•**Language**: Mathematics and technical vocabulary

•**Tools**: uses instruments and technologies

• **Process**: Organized, but very diverse activity shaped by social forces and historical change

Technology…

• A discourse or treatise on an art or arts; The scientific study of the practical or industrial arts. Techne (art, craft, skill), Logos (word). a. b. c. Artifacts or Hardware. **Products fabricated by humans to meet specific needs.** Tools, machines, implements. Knowledge and Methods. A system of tacit and explicit knowledge, techniques, and materials utilized in using, making, or repairing a certain kind of artifact. A human cultural activity or profession. e.g. military or civil engineers, crafters, machinists. **It is an application of scientific knowledge to the practical aims of human life to change and manipulate the human environment.**

Impacts on the society

•The shift from nomadic life to farming led to the development of cities

•Network of transportation, communication and trade systems

•Technical innovations saved physical energy and lessened people's workload

• **The greatest innovation of technology was longevity**

•A large part of technology has been dedicated to the advancement of medical science

Scientific revolution-was the period of enlightenment when the development in the field of mathematics, physics, astronomy, biology, and chemistry

transformed the views of society about nature.

HISTORICAL ANTECEDENT IN WHICH SOCIETY CONSIDERATION CHANGE THE COURSE OF SCIENCE AND TECHNOLOGY.

The Copernican Revolution: Beginning of Modern Astronomy: The 15th century Astronomy

• The knowledge about the **nature of the universe** had been unchanged **since the Greek antiquity.** Most astronomers believed the theory of Greek astronomer **Ptolemy** which had developed more than a millennium earlier. The Church during that time adhered to the idea of **Geocentrism** and that **anyone who opposes is considered heretic thus, subject to inquisition.**

• Most astronomers believed the theory the Greek astronomer Ptolemy had developed more than 1,000 years earlier. Ptolemy said the Earth was the center of the universe and was motionless. He believed all other heavenly bodies moved in complicated patterns around the Earth.

• **Copernicus felt that Ptolemy's theory was incorrect**. Sometime between **1507 and 1515**, he first circulated the principles of his **heliocentric or Sun-centered astronomy.**

Two competing models proposed

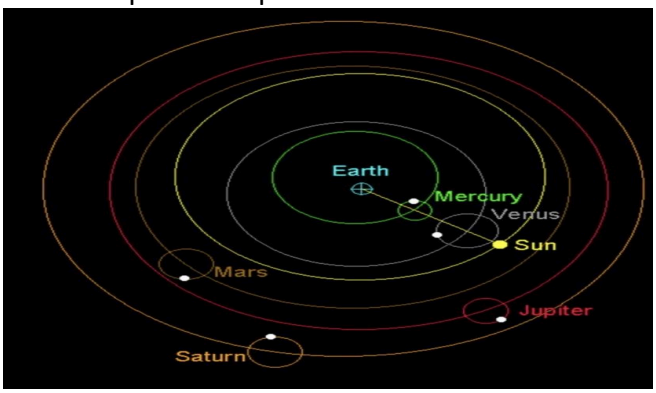
1. **Geocentrism**

The word "**geo**" in Greek means "**earth**." **Geocentric means that the earth is in the center**. In the geocentric system, the earth is statically located at the center and the rest of the planets revolve around it, including the sun, which was also considered a planet.

•The planet nearest to earth is the moon, which completes a single revolution about the earth in one month. Further out are the rest of the planets: Mercury, Venus, the Sun, Mars, Jupiter and Saturn.

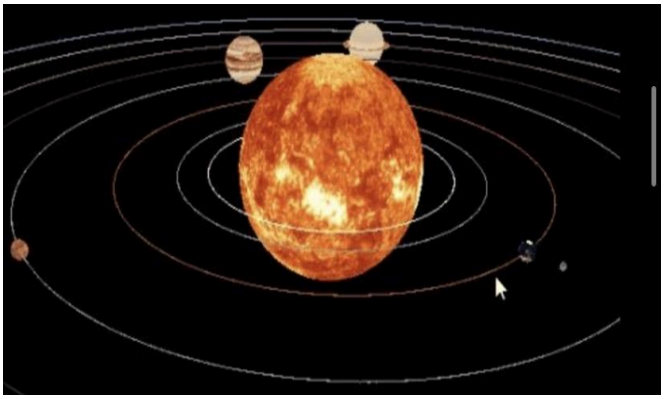
• **InPtolemaic** model, planets move in "**epicycles**" which are circles concentric to another circle around the Earth called "deference". Hence, planets follow a uniform circular motion.

• Aristotle, also established a geocentric universe in which the fixed spherical earth is at the center surrounded by concentric celestial spheres of planets and star.

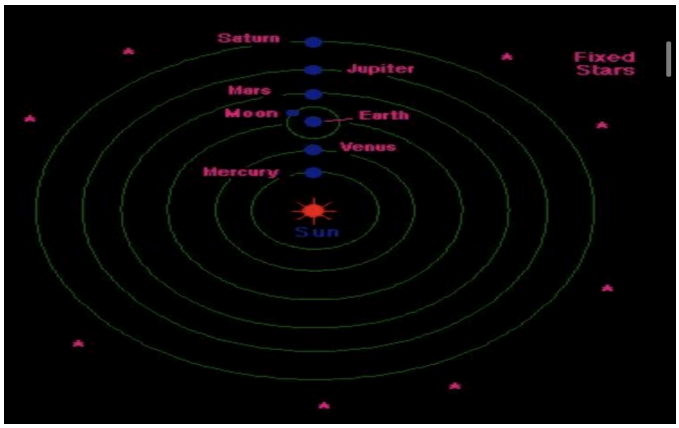


2. **Heliocentrism**

•The Heliocentric Theory is the astronomical model in which **the Earth and planets revolve around a relatively stationary Sun at the center of the Solar System.** The word "**helios**" in Greek means "**sun**." Heliocentric means that the sun is at the center.



• The Earth and all the planets are orbiting around the Sun positioned into an increasing sequence of distance. Therefore, Copernicus claims that the orbits of Venus, Mercury are closer to the Sun while Mars, Saturn and Jupiter are farther to the Sun.



Key concepts of the Copernican Revolution

• Polish astronomer **Nicolaus Copernicus** (1473 – 1543) attested the idea that the Sun is the center of the universe where planets revolve around it while other celestial bodies such as the moon orbit the Earth. In an important way, Copernicus bears a resemblance to ancient Greek Philosophers as he did not do extensive experiments. Instead, everything came from his own ideas or thinking which was referred to as **thought experiment.**

• It is believed by many that his book was only published at the end of his life because he **feared ridicule and disfavor by his peers and by the Church**, which had elevated the ideas of Aristotle to the level of religious dogma.

The publication of his book, "**On the revolutions of the Heavenly spheres"** in the early **16th century** is often cited as the beginning of the **scientific revolution.**

Points of the Copernican Model

The celestial spheres do not have one common center. The Earth is not at the center of everything. Earth is not the center of the universe, only the center of gravity and the lunar orbit. Only the Moon orbits Earth. All the spheres orbit the Sun. Spheres means the planets. Compared to the distance to the stars, the Earth to Sun distance is almost nonexistent. The stars are very much farther away than the Sun. The motion of the stars is due to the Earth rotating on its axis; The retrograde (clockwise) and forward motions of planets is caused by the Earth's motion. The motion of the Sun is the result of the Earth's motions. (rotation and revolution).

Copernicus' Influence on Later Scientists

Tycho Brahe's Observations

• Developed a comprehensive method of recording precise observations of the skies, before the advent of the telescope. His many detailed observations included:

The discovery of a supernova (a new star, to all appearances, which shone brightly and then faded away) in the heavens, in 1572

Tycho Brahe's Observations (1546 –1601)

• Tycho proposed an intermediate model of the universe, where the Earth was still the center point, but the other planets were allowed to rotate around the Sun as it rotated around the Earth. [Tychonic system]

• Brahe believed in a model of the Universe with the Sun (rayed disk) orbiting the Earth (black dot), but the other planets (symbols) orbiting the Sun. In an attempt to prove his

theory, Brahe compiled extensive astronomical records.

Johannes Kepler

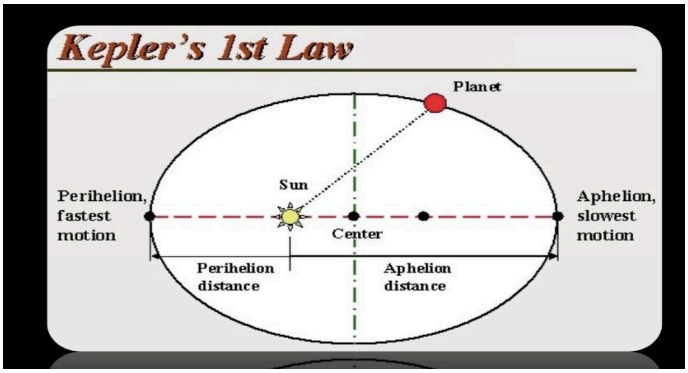
•While Copernicus rightly observed that the planets revolve around the Sun, it was Kepler who correctly defined their orbits.

•From the researches of Tycho Brahe, Kepler use it to prove the Copernican theory after the death of Brahe.

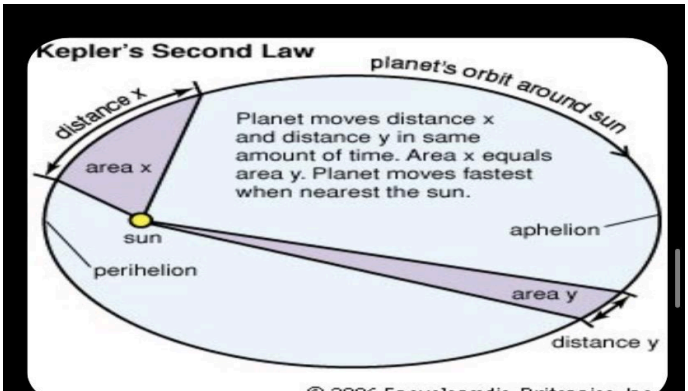
•Kepler eventually proved heliocentrism and calculated the orbital laws. He found that the orbits of the planets followed three laws.

Johannes Kepler's Laws of Planetary Motion

1st Law: Law of Ellipse: that the planets move in an ellipse (a squashed circle) with the Sun at one focus point, offset from the center.

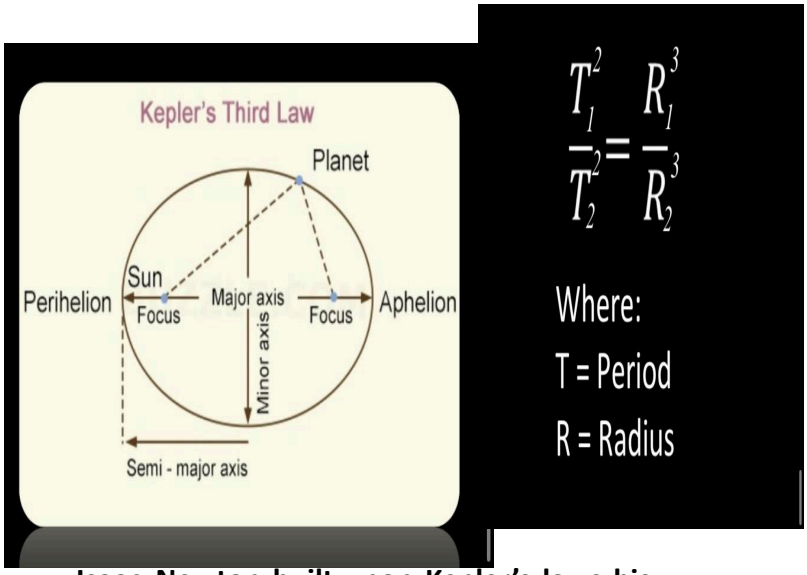


2nd law: Law of Equal Areas : the planet must move more quickly when it is near the Sun, but more slowly when it is farthest from the Sun• 3rd law: Law



• 3rd law: Law of Periods: there is a precise mathematical relationship between a planet's distance from the Sun and the amount of time it takes revolve around the Sun. It was this law that inspired Newton, who came up with three laws of his own to explain why the planets move as they do.

Period is the time it took a planet to revolve around the Sun; Radius is the average distance from the Sun. Hence, the orbits of planets are essentially equal.



Isaac Newton built upon Kepler's laws his Law of Gravitation

•If it wasn't for Mars and its travels across the system, Johannes may not have developed his laws and consequently, Newton's formulation of his laws as well;

•Kepler's laws provide accurate evidences to Newton's laws of motion and universal gravitation

Galileo Galilei (1564 – 1642)

•Individual stars within the Milky Way

• Phases for Venus, like those of the Moon

• Moons orbiting Jupiter (Io, Europa, Ganymede and Callisto)

• Mountains and volcanoes on the Moon

• Sunspots

CHARLES DARWIN (1809-1882)

BIRTH AND FAMILY

Born on February 12, 1809 in Shrewsbury, England.

Second to the youngest child of the wealthy and influential family of scientists whose father is Robert Waring Darwin, a medical doctor who also desired Darwin to follow his footsteps in becoming a medical doctor. Thus, his mother Susana died when he was at the early childhood age of 8.

EDUCATION

In 1825, Darwin enrolled at University of Edinburgh to study medicine but also left after 2 years as the sight of blood made him queasy. Then, transferred to Cambridge University wherein after completing his Bachelor of Arts degree in 5 years he was offered a position of being a naturalist in HMS Beagle. Whereby, this voyage proved an opportunity to Darwin who's inclined in the study of natural history

PERSONAL DETAILS

A naturalist who is well known in the development of evolution in explaining biological change. Charles Darwin proposed then married to his first cousin, Emma Wedgewood on January 29, 1839 and had ten (10) children where three (3) of them died in childhood.

He was awarded a Copley Medal, the greatest honor in Science.

Darwin began to suffer from chronic illnesses which occurred after returning from his trip around the world. Having contracted Chagas disease, a parasitic illness which resulted in cardiac damage and caused Darwin's death on April 19, 1882 at his country house.

He was buried at Westminster Abbey, London.

RELIGIOUS VIEWS

Darwin argued, "how had this come about? Could this extraordinary variety really be explained by the idea that God had created all the species on Earth in six days?"

He worked on these questions over the years and concluded that offspring from any species resembled their parents but also varied from them.

His theory offended many for it meant that there was no need for a creator-God and it placed mankind firmly within the animal kingdom.

Though he was probably an agnostic, he was always polite to and about religious believers.

HUMAN SOCIETY

On Darwin's second book, The Descent of Man, he then stated that "Ultimately a highly complex sentiment, having its first origin in the social instincts, largely guided by the approbation of our fellow men, ruled by reason, self-interest, and in later times by deep religious feelings, confirmed by instruction and habit, all combined,

constitute our moral sense or conscience," as he observed social feelings in animals and assumed that it could be a natural and instinctive in humans, putting forward the idea that morality existed before religion

THE VOYAGE OF THE BEAGLE

In 1831, Charles Darwin was just 22 years old when he set sail on a circumnavigation of the globe on a ship called HMS Beagle.

He was the naturalist on the expedition and it was his job to collect and observe specimens of plants, animals, rocks, and fossils when the ship went ashore.

Darwin made many observations during the voyage that helped him formulate his Theory of Evolution such as:

Exploring plants and animals he had never seen before in different tropical rainforests and other new habitats, thus impressed him with the great diversity of life;

Experiencing an earthquake that elevated the ocean floor 2.7 meters (9 feet) above sea level wherein he also found rocks containing fossil sea shells in mountains high above sea level, hence suggesting that oceans and continents had intensely reformed over time and still continue to change in drastic ways

CHARLES LYELL

Visiting rock ledges that had once been beaches that had gradually built over time which implied that slow and steady processes also change the Earth's surface;

Digging up fossils of gigantic extinct animals like ground sloth which served as proof that organisms looked very different in the past and suggested that living things also change over time.

Darwin's most important observations were made on Galapagos Islands.

He observed different birds and animals that had evolved in various environments which led him to form the theory of natural selection.

The theory states that variations within species occur randomly and the adaptability of an organism determines its survival or extinction.

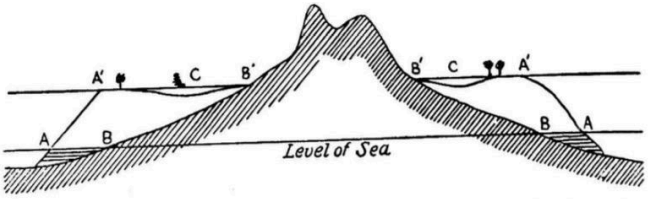
Alfred Russell Wallace, another young naturalist in that time, had independently come to similar conclusions about evolution and natural selection.

FORMATION OF CORAL REEFS AND ATOLLS

During the voyage, Darwin also made significant observations about the geology of the islands and coastlines he visited and proposed a theory about the Formation of Atolls.

Atolls, sometimes called coral atolls, are coral reefs formations that encircle a lagoon. They are mostly found in the Pacific and the Bikini Atoll located in the northwest of Hawaii is an example.

According to Darwin, a fringing reef would form from corals growing along the edges of a newly formed volcanic island

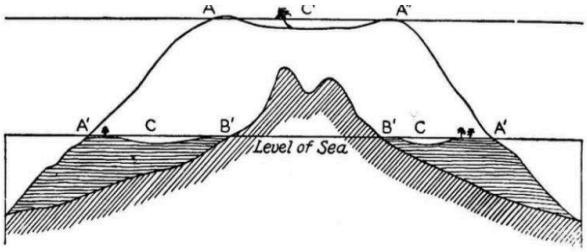


AA. Outer edges of the fringing-reef, at the level of the sea. BB. The shores of the fringed island.

A'A'. Outer edges of the reef, after its upward growth during a period of subsidence, now converted into a barrier, with islets on it B'B'. The shores of the now encircled island. CC. Lagoon-channel.

N. B.—In this and the following woodcut, the subsidence of the land could be represented only by an apparent rise in the level of the sea.

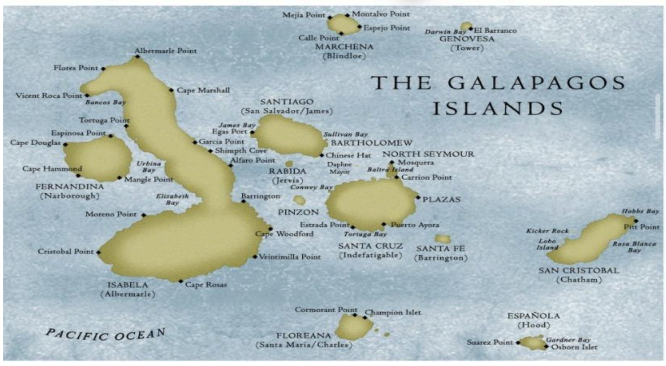
A lagoon would then open between the island and the reef when the island began to sink, thus forming a barrier reef. Then, the island would eventually disappear, leaving an atoll

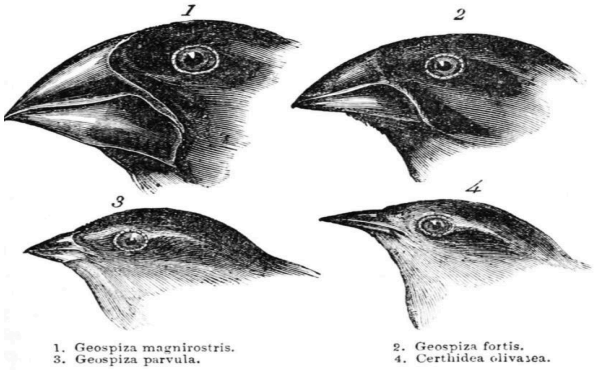


A′A′. Outer edges of the barrier-reef at the level of the sea, with islets on it. B′B′. The shores of the included island. CC. The lagoon-channel.
A″A″. Outer edges of the reef, now converted into an atoll. C′. The lagoon of the new atoll.
N. B.—According to the true scale, the depths of the lagoon-channel and lagoon are much exaggerated.

To give sustenance to the theory, Darwin explored and found dead corals on the lagoon bottoms below 120-foot limit. This finding was inferred to confirm his proposition about the subduction of sea bed around volcanic islands.

ON THE ORIGIN OF SPECIES





1. Geospiza magnirostris.
2. Geospiza fortis.
3. Geospiza parvula.
4. Certhidea olivacea.

[The title page and facing quotations of a first edition of Origin of species on loan to Christ's College, Cambridge. Photograph: **John van Wyhe.**

DARWIN AND THE FIT- Charles Darwin published On the Origin of Species which included the oft misquoted line 'survival of the fit.' Fit, not fittest, was originally written.

The idea here was that one merely needed to be fit enough to reproduce to pass on its genetics in the form of a new generation

SPENCER AND THE FITTEST

Herbert Spencer was a 19th-century philosopher and social idealist.

Born on April 27, 1820 and died on December 8, 1903.

He was known to be a strong supporter of Darwin's methods of evolution and a supporter of the evolutionary process.

Herbert Spencer's works modified the line, making it 'survival of the fittest.' This idea and quote were then used as the foundation for what would become known as Social Darwinism.

SOCIAL DARWINISM

Survival of the fittest implies that the strong will succeed and the weak shall perish.

The 'fittest' will be successful and they shall rule the weaker because they are the most fit to do so.

In a brutal world without social hierarchy, class, and social

customs, this would mean the person who is the strongest and has the longest spear rules over everyone.

Social Darwinism is the application of one interpretation of the law of evolution; those who succeed in society were superior to those who did not succeed

EVOLUTION

Described as “descent with modification”.

It is the idea that species change over time, give rise to a new species and share a common ancestor.

Each species has its unique set of heritable genetic differences from the common ancestor by which can split-off producing a multi-level tree linking of all organisms

EVIDENCES OF EVOLUTION

1. Fossil Record

Fossils are the preserved remains of previously living organisms or their traces, dating from the distant past. Fossils document the existence of now-extinct species, showing that different organisms have lived on Earth during different periods of the planet's history. They can also help scientists reconstruct the evolutionary histories of present-day species.

2. Comparative anatomy is the study of the

similarities and differences in the structures of different species. Similar body parts may be homologous structures or analogous structures.

Homologous structures are structures that are similar in related organisms because they were inherited from a common ancestor. These structures may or may not have the same function in the descendants. Analogous structures are structures that are similar in unrelated organisms. The structures are similar because they evolved to do the same job, not because they were inherited from a common ancestor. Vestigial structures-reduced forms of functional structures.

3. **Comparative embryology** is the study of the similarities and differences in the embryos of different species.

4. Comparative Biochemistry

Biochemistry is the study of the basic chemistry and processes that occur in cells. It involves the comparison of DNA, proteins such as cytochrome c and hemoglobin and chromosomes across different species. Similarities in the comparison can be used as evidence in support of evolution.

Comparing Proteins

Different species of organisms may contain the same genes with slight variations in sequences. For example, the coding for cytochrome c is very similar in humans, chickens, mice and horses.

5. Evidence from Biogeography

Biogeography is the study of how and why organisms live where they do. It provides more evidence for evolution.

Darwin believed that there is a process by which organisms that can better adapt in their environment do survive, while those less able to adapt become extinct. This process is called?**Adaptation**

NATURAL SELECTION

It is the process by which organisms better suited to adapt in their environment survive, while those less able to adapt become extinct.

The result is not a sudden creation of each individual species but a successive evolution and improvement of a species

Key Components of Natural Selection

A. Overpopulation

The number of offspring is usually greater than the available resources necessary for organisms to survive.

Thomas Robert Malthus- Population grows geometrically while food production grows arithmetically.

Organisms can die from many causes: disease, starvation, or being eaten by predator

B. Struggle for Existence

Every organism must struggle to survive.

Organisms must compete for limited resources, including food, territory, and other necessities of life to survive.

C. Presence of Variation

There is variation within a species.

Species is a group of organisms that share similar characteristics and interbreed to produce fertile offspring. However, with the advent of molecular cloning, it is now possible to artificially interbreed two different species

D. Survival of the Fittest

Some of variations allows members of a species to survive and reproduce better than others.

Certain traits help organisms survive and reproduce in a given environment

E. Variations can be transmitted

Organisms that survive and reproduce gradually transmits their traits to their offspring creating more and more of the population.

THE DESCENT OF MAN AND SELECTION IN RELATION TO SEX

The second book of Charles Darwin following his main book On the Origin of the Species.

The book discusses about the relationship of the Theory of Evolution and the Theory of Sexual Selection.

He wrote this book to explore the following concepts: first, is whether a man is descended from some pre-existing form, second, the manner of that development and lastly, the value of the differences between the so-called races of man.

The book is mainly divided into two volumes namely: The Evolution of Man and Sexual Selection

THE EVOLUTION OF MAN

He tackles evidences to argue that humans evolved from lower form of animal.

He examines the similarities between man and some higher animals like primates, dogs, horses and other mammals.

He began to compare their anatomy and brain capability.

He also explored the similarity of some of their characteristics and behavior corresponding on how they adapt

Charles Darwin then continues his work by further by explaining the role of Natural Selection and Genetic Inheritance, stating that both humans and animals have traits and intellectual ability which can be passed down through generations that can remain constant or can change overtime.

Ending the first part of the book, he rejected both idea of monogenism and polygenism, stating that a possible reason for the divergence of racial differences was caused by sexual selection.

SELECTION IN RELATION TO SEX

Charles Darwin proposed the idea of sexual selection, in which states that a female chooses a mate with a more attractive appearance and a more dominant trait, to explain why there is a physical feature of some animals with no function. This implies that animals also evolve to gain reproductive advantage.

He also argues that, through constant developing of males, the choosiness of the females has made them inferior to the males.

For more complex organisms, more factors are considered in choosing a mate. Charles Darwin connected this idea with the culture of sexual partnership within human nature.

Thus, he concluded that humans can further improve their condition through the struggle for their existence.

Who is Sigmund Freud?

An Austrian Neurologist and founder of Psychoanalysis

The Case of Anna O

TRUE NAME: Bertha Pappenheim.

Anna O. was a pseudonym

Patient of Dr. Josef Breuer

Her case was described in the book that Breuer wrote with Sigmund Freud, Studies on Hysteria.

Anna O. came to Josef Breuer for treatment for what was then known as hysteria. While caring for her dying father, Pappenheim experienced a range of symptoms that included partial paralysis, blurred vision, headaches, and hallucinations.

During the treatment, talking about Anna's experiences seem to offer some relief from her symptoms that was later found out by Breuer. She dubbed the treatment the "talking cure."

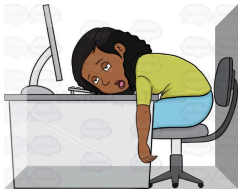
Freud never actually met Anna O. Anna O's story fascinated Freud and it served as a basis for Studies on Hysteria. Breuer's description of her treatment led Freud to conclude that hysteria was rooted in childhood sexual abuse.

"LATENCY INCUBATION"



HYDROPHOBIA





LETHARGY

LANGUAGE DIFFICULTIES



Id:
Instincts



Ego:
Reality

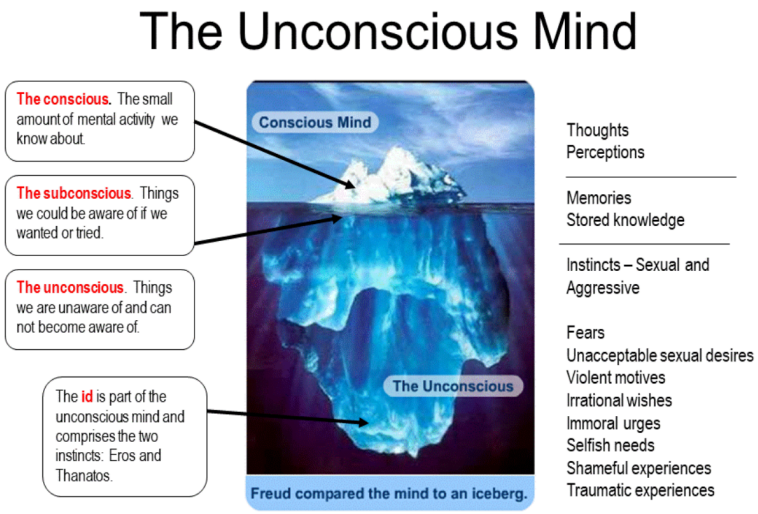


Superego:
Morality

ANNA'S TREATMENT Talking therapy

PSYCHOANALYSIS is a set of theories and therapeutic techniques related to the study of the unconscious mind, which together form a method of treatment for mental-health disorders.

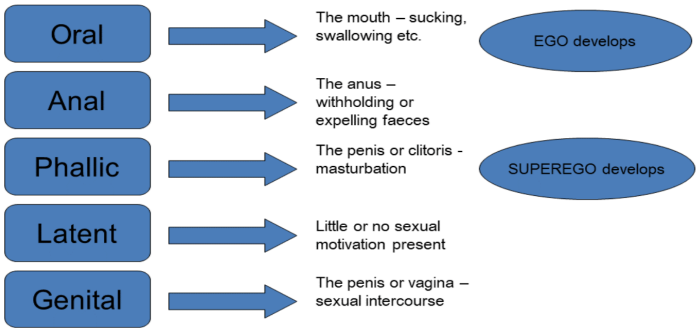
Personality is formed through conflicts among three fundamental structures of the human mind: the id, ego, and superego.



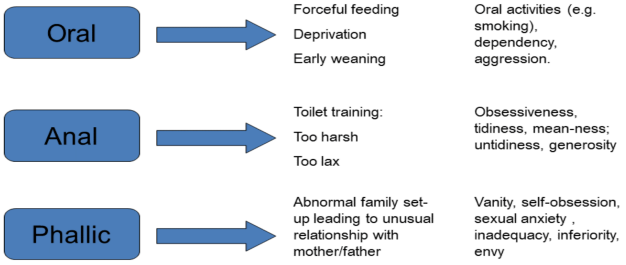
Structural Model of Mind

Freud (1923) later developed a more structural model of the mind comprising the entities id, ego, and superego (what Freud called "the psychic apparatus"). These are not physical areas within the brain, but rather hypothetical conceptualizations of important mental functions.

PSYCHOSEXUAL STAGES



Fixation



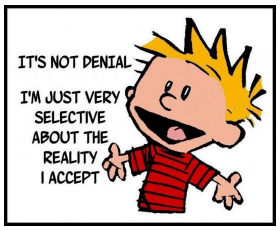
To be psychologically healthy, we must successfully complete each stage. Mental abnormality can occur if a stage is not completed successfully and the person becomes 'fixated' in a particular stage. This particular theory shows how adult personality is determined by childhood experiences.

DREAM ANALYSIS

Freud (1900) considered dreams to be the *royal road to the unconscious* as it is in dreams that the ego's defenses are lowered so that some of the repressed material

comes through to awareness, albeit in distorted form. Dreams perform important functions for the unconscious mind and serve as valuable clues to how the unconscious mind operates.

DEFENSE MECHANISM



Mechanism	Description	Example
Repression	Repression is an unconscious mechanism employed by the ego to keep disturbing or threatening thoughts from becoming conscious.	During the Oedipus complex aggressive thoughts about the same sex parents are repressed
Denial	Denial involves blocking external events from awareness. If some situation is just too much to handle, the person just refuses to experience it.	For example, smokers may refuse to admit to themselves that smoking is bad for their health.
Projection	This involves individuals attributing their own unacceptable thoughts, feeling and motives to another person.	You might hate someone, but your superego tells you that such hatred is unacceptable. You can 'solve' the problem by believing that they hate you.
Displacement	Satisfying an impulse (e.g. aggression) with a substitute object.	Someone who is frustrated by his or her boss at work may go home and kick the dog,
Regression	This is a movement back in psychological time when one is faced with stress.	A child may begin to suck their thumb again or wet the bed when they need to spend some time in the hospital.
Sublimation	Satisfying an impulse (e.g. aggression) with a substitute object. In a socially acceptable way.	Sport is an example of putting our emotions (e.g. aggression) into something constructive.

Ritual and undoing- Doing the same action again and again to cancel out a bad thought or feeling, or to stop something bad from happening. The person feels they must do it, even if it makes no real sense.
Example: A person feels guilty about an angry thought, so they keep washing their hands. Or someone checks the door lock three times to keep the house safe.
Reaction formation-Showing the opposite of what you truly feel. The person hides a feeling they cannot accept by acting in the reverse way, often too much.
Example: A mother who secretly does not want her child becomes too caring and spoils the child.

rationalization-Giving a good-sounding reason for something to hide the real reason. It protects the person from feeling bad.*Example:* A person who gets fired says, "I wanted to quit anyway."

NEO-FREUDIANS

Alfred Adler



- The first to explore and develop a comprehensive social theory of the psychodynamic person.
- Proposed the concept of the inferiority complex, which describes a person's feelings that they lack worth and don't measure up to the standards of others or of society.
- Believed in the importance of social connections, seeing childhood development as emerging through social development rather than via the sexual stages outlined by Freud. From these ideas, he identified three fundamental social tasks that all of us must experience: *occupational tasks (careers), societal tasks (friendship),* and *love tasks (finding an intimate partner for a long-term relationship.*

Erik Erikson



- Proposed the psychosocial theory of development, which suggests that an individual's personality develops throughout the lifespan based on a series of social relationships.
- In his psychosocial theory, he emphasized the social relationships that are important at each stage of personality development.
- He identified eight stages, each of which represents a conflict or developmental task. The development of a healthy personality and a sense of competence depend on the successful completion of each task.

Carl Jung



- He developed a theory of personality called analytical psychology.
- One of his major contributions was his idea of collective unconscious, which he deemed a "universal" version of Freud's personal unconscious, holding mental patterns, or memory traces, that are common to all of us.
- He also proposed the concept of the persona, referring to a kind of "mask" that we adopt based on both our conscious experiences and our collective unconscious.

Questions and Answer

Charles Darwin came from what kind of family?? **Family of Scientists**

What position was offered to Darwin after completing his Bachelor of Arts degree? **Naturalists**

What is the title of Darwin's book that was published on Nov. 24 1859, and is the foundation of evolutionary biology? **On the Origin of Species**

What was Darwin's religious view before he dwelled on his research? **Christian**

What is the second book of Charles Darwin that speaks up about his views and opinions to the human society? **The Descent of Man**

It is the island where Darwin's most important observations were made on. **Galapagos Island**

Darwin proposed a theory about coral reefs formation that encircles a lagoon called **Atolls**

What species (it displayed different structural adaptations such as its different-shaped beaks) does Darwin found on the Galapagos Island?**Finches**

The theory that can be described as "descent with modification" and one of the substantiated theories in the history of Science. **Evolution by Natural Selection**

In “The Descent of Man”, Charles Darwin expands on ideas that he first expressed. **The Expression of Emotions in Man and Animal**

The theory that human race has ascended from two or more ancestral types. **Polygenism**

According to Charles Darwin, what do men have to offer in order for them to be more valuable to women as their mate?. **Attractive appearance**